

The maximum number of stable matchings of order six is forty-eight

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Abstract

Let $f(n)$ denote the maximum number of stable matchings of a stable-marriage instance with n men and n women (Knuth 1976; Gusfield–Irving Open Problem #1). The known values were $f(1), \dots, f(5) = 1, 2, 3, 10, 16$; the value of $f(6)$ was open, with the lower bound 48 (a dihedral Latin instance) conjectured exact (OEIS A357271). We prove $f(6) = 48$. The proof is a reduction of the $\sim 10^{28}$ -instance search space to a finite space of *rotation schedules*: sequences of cyclic partner exchanges applied from the identity matching. Three elementary lemmas over the standard rotation theory show that every instance is dominated, in stable-matching count, by the *read-off instance* of its own schedule, and that a first-appearance labeling and a commutation normal form lose no schedules. An exhaustive enumeration then evaluates the exact stable-matching count of all 26,574,282,886 schedule nodes — roughly ten hours on a laptop — attaining maximum 48. The methodology is validated against ground truth: run at orders 3, 4, 5 it reproduces 3, 10, 16, the last being the value the author’s companion development certifies formally (Lean + verified SAT certificates); an independent implementation and canonicalization on/off comparisons (up to a 2.66×10^9 -node check) agree throughout. The bridge and validity lemmas are machine-checked in Lean 4 (no `sorry`, standard axioms); the symmetry lemma is validated computationally, and a verified replay of the enumeration remains future work.

1 Introduction

A stable-marriage instance of order n equips n men and n women each with a strict preference order over the opposite side; a perfect matching is *stable* if no man and woman prefer each other to their assigned partners. Knuth [1] asked for the behaviour of $f(n)$, the maximum number of stable matchings over instances of order n ; Gusfield and Irving made it Open Problem #1 of their monograph [2]. Asymptotically $2.28^n \leq f(n) \leq 3.55^n$ [3, 4, 5]. Exact values have been scarce: $f(4) = 10$ and $f(5) = 16$ are due to Eilers [6] (the latter obtained in 2022 by constraint programming and certified formally in the author’s companion work [7]), and

$$f(1), \dots, f(5) = 1, 2, 3, 10, 16.$$

For $n = 6$ the best lower bound has long been 48, attained by a Latin instance built from the dihedral group; OEIS A357271 records the conjecture that 48 is exact. The instance space of order six ($\sim 10^{28}$ after obvious symmetries) is far beyond direct search, and the best human-provable upper bound, $3.55^6 \approx 2005$, is forty times too large.

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Result and method. We prove $f(6) = 48$. The key is a change of universe: instead of instances, we enumerate *schedules* — sequences of cyclic partner exchanges (“rotations”) applied from the identity matching, subject to the budget constraints that rotation theory imposes (at most 30 moved-man slots, at most 5 moves per man, no partner revisited). Each schedule *reads off* an instance whose preference lists are the participants’ partner trajectories, completed at the bottom. Three lemmas (Section 3) show: (i) every instance’s rotations, in elimination order, form a valid schedule; (ii) the read-off instance of that schedule has at least as many stable matchings (the *bridge lemma*); and (iii) a first-appearance labeling rule and a commutation normal form preserve at least one representative of every schedule, without affecting read-off counts. The schedule space is small enough to exhaust: our enumerator visits 26,574,282,886 nodes and computes the exact stable-matching count of every node’s read-off instance, attaining maximum 48 (Section 5). With the dihedral lower bound, $f(6) = 48$ follows.

Validation. Because the result rests on an enumeration, Section 7 documents the checks: the same machine run at orders 3, 4, 5 reproduces $f = 3, 10, 16$ — the last value being formally certified (machine-checked in Lean, with SAT refutations verified by the formally verified checker `cake_lpr`) in the companion work [7], so a certified theorem serves as test oracle for the present method; an independently written enumerator agrees on maxima; disabling the symmetry reductions multiplies the tree by up to $48\times$ (a 2.66×10^9 -node check at order six) and never changes a maximum; and 1.3×10^9 local-search evaluations in two other search spaces also never exceeded 48. Section 8 describes the Lean 4 formalization of the reduction.

2 Preliminaries

We use the structure theory of the stable-matching lattice and its rotations (Gusfield–Irving [2], Chapters 2–3) and record what we need.

Fact 1. *Call (m, w) a stable pair of an instance I if some stable matching contains it. For each man m , his stable partners ordered by his preference form his trajectory T_m : the man-optimal matching μ_0 gives each man the first entry of his trajectory, and the rotations of I , applied from μ_0 in any linear extension of the rotation poset, move each man strictly down his trajectory, one entry per rotation containing him, ending at the woman-optimal matching. Symmetrically each woman moves strictly up her trajectory. A rotation is a cyclic sequence $(m_0, w_0), \dots, (m_{k-1}, w_{k-1})$, $k \geq 2$, of pairs of the current matching, whose application gives m_i the partner w_{i+1} of m_{i+1} (indices mod k). The number of stable matchings equals the number of downsets of the rotation poset.*

Fact 2. *Each man is moved by exactly $|T_m| - 1 \leq n - 1$ rotations, so the total move count satisfies $\sum_\rho |\rho| = \sum_m (|T_m| - 1) \leq n(n - 1)$; for $n = 6$: at most 30 slots, at most 5 moves per man, and every rotation moves at least two men.*

3 The schedule reduction

Definition 1 (Schedule). *Fix $n = 6$ and start from the identity matching. A schedule is a sequence of steps, each a cyclic move on $k \geq 2$ men (man m_i takes the current wife of m_{i+1}), subject to: total move count ≤ 30 , at most 5 moves per man, and no man ever moving to a wife he has already had, nor any woman receiving a husband she has already had.*

Definition 2 (Read-off instance). *Given a schedule S , the read-off instance $R(S)$ ranks, for each man, his trajectory in order at the top of his list, followed by the remaining women in a fixed canonical order; for each woman, her trajectory reversed at the top, followed by the remaining men canonically.*

Lemma 1 (Validity). *For any instance I of order 6, relabel women so that the man-optimal matching is the identity. Then the rotations of I , in any linear extension of the rotation poset, form a schedule $S(I)$.*

Proof. Rotations are cyclic moves on $k \geq 2$ men exposed in the current matching, and applying them in a linear extension realizes the lattice traversal, in which men move strictly down and women strictly up their trajectories (hence no repeats). The caps are Fact 2. $\square$

Lemma 2 (Bridge). *For every instance I of order 6 (women relabelled as above), $\text{sc}(I) \leq \text{sc}(R(S(I)))$.*

Proof. Write $J = R(S(I))$. The trajectories of $S(I)$ are the stable-partner sequences of I (Fact 1); thus for each man, J ranks his I -stable partners in the same relative order as I does, with all other women strictly below, and symmetrically for women. We claim every I -stable matching μ is J -stable, which proves the inequality. Every pair of μ is a stable pair, hence a trajectory pair on both sides. Let (m, w) , $w \neq \mu(m)$. If $w \notin T_m$: in J , m ranks w below all of T_m , in particular below $\mu(m)$; no block. If $w \in T_m$: then (m, w) is a stable pair, so $m \in T_w$; also $\mu(m) \in T_m$ and $\mu^{-1}(w) \in T_w$. Since J preserves relative I -order within trajectories on both sides, (m, w) blocks μ in J iff it blocks in I — and it does not. $\square$

Lemma 3 (Symmetry soundness). *(i) Exchanging two adjacent steps with disjoint man-sets leaves all trajectories, hence $R(S)$, unchanged. (ii) Relabelling men and women by a common permutation maps schedules to schedules and read-off instances to relabelled instances of equal count. Consequently an enumeration that (a) requires the new men of each step to be exactly the next unused labels as a set, and (b) rejects a step lex-smaller than an earlier step separated from it only by man-disjoint steps, still attains $\max_S \text{sc}(R(S))$.*

Proof. (i) Steps with disjoint man-sets touch disjoint women (their women are their men's current wives). (ii) Immediate. For the consequence, relabel by first participation (any assignment within a step's new set satisfies the set rule), and note rule (b) discards a sequence only in favour of a commutation-equivalent one with the same read-off instance. $\square$

4 A worked example

Order 3 is small enough to display the entire reduction. Consider the cyclic Latin instance I_3 (each row lists partners from most to least preferred):

$$\begin{array}{c|ccc} m_0 & w_0 & w_1 & w_2 \\ m_1 & w_2 & w_0 & w_1 \\ m_2 & w_1 & w_2 & w_0 \end{array} \qquad \begin{array}{c|ccc} w_0 & m_2 & m_1 & m_0 \\ w_1 & m_1 & m_0 & m_2 \\ w_2 & m_0 & m_2 & m_1 \end{array}$$

It has $f(3) = 3$ stable matchings. In the man-optimal one every man gets his first choice: $\mu_0 = \{m_0w_0, m_1w_2, m_2w_1\}$; the woman-optimal one gives every woman hers. The rotation

poset is a 2-chain of two rotations, each moving all three men, so the number of downsets is 3, as it must be.

Relabel women so that μ_0 becomes the identity ($w_0 \mapsto 0, w_2 \mapsto 1, w_1 \mapsto 2$). The two rotations become two applications of the same cyclic step ($m_0 m_2 m_1$) — written (021) in min-first form — taking the matching through

$$(0, 1, 2) \longrightarrow (2, 0, 1) \longrightarrow (1, 2, 0).$$

Man m_0 's trajectory is (w_0, w_2, w_1) , which is exactly his (relabelled) preference order: at order 3 every pair of I_3 is a stable pair, so the trajectories exhaust the preference lists, the bottom completion is empty, and the read-off instance $R(S(I_3))$ is I_3 itself — the instance is a fixed point of the reduction, as every Latin extremal instance is.

The entire order-3 schedule space, enumerated under Definition 1 with the reductions of Lemma 3, has eight nodes:

schedule	$sc(R(S))$	schedule	$sc(R(S))$
ε	1	(012)	2
(01)	2	(012)(012)	3
(01)(02)	3	(021)	2
(01)(12)	3	(021)(021)	3

The maximum is $3 = f(3)$, attained four times; $S(I_3) = (021)(021)$ is among the attainers, as Lemma 1 promises. Note that a maximizer need not use large steps — (01)(02), two transpositions, already attains it — a first glimpse of the all-size-2 phenomenon of Section 9.

5 The enumeration

The enumerator (`gen_enum.c`, ~200 lines of C) performs a depth-first search over schedules. The steps available at order 6 are the

$$\sum_{k=2}^6 \binom{6}{k} (k-1)! = 15 + 40 + 90 + 144 + 120 = 409$$

cyclic moves on 2..6 men in min-first form; state comprises the current matching, per-person trajectories and visit masks; the reductions of Lemma 3(a,b) and the caps of Definition 1 prune the tree. At *every* node the read-off instance is built (trajectory prefix + bottom completion) and its stable-matching count computed exactly by scanning all 720 perfect matchings. Sharding on the depth-1 step splits the work across cores. The same machine, parametrized by order, produces:

order	step shapes	nodes	$\max_S sc(R(S))$	$f(n)$
3	5	8	3	3
4	20	409	10	10
5	84	498,599	16	16
6	409	26,574,282,886	48	48

The full run (512 shards, 8 cores of an Apple M4 Max, ~ 10 hours) visits 26,574,282,886 nodes, evaluates all of them, and attains

$$\max_S \text{sc}(R(S)) = 48.$$

Because every node is evaluated, no monotonicity assumption about schedule extension is needed: for any instance I , the node $S(I)$ itself is evaluated (Lemma 3).

Theorem 1. $f(6) = 48$.

Proof. The dihedral Latin instance attains 48 (direct computation, three independent counters; OEIS A351413). Conversely, for any instance I , $\text{sc}(I) \leq \text{sc}(R(S(I))) \leq 48$ by Lemmas 1–3 and the enumeration. $\square$

6 The extremal instance

The unique-up-to-symmetry maximizer known at order 6 is Eilers’ Latin instance built from the dihedral group (OEIS A351413):

m_0	w_0	w_1	w_2	w_3	w_4	w_5	w_0	m_5	m_4	m_3	m_2	m_1	m_0
m_1	w_1	w_0	w_3	w_2	w_5	w_4	w_1	m_2	m_3	m_4	m_5	m_0	m_1
m_2	w_4	w_3	w_0	w_5	w_2	w_1	w_2	m_3	m_2	m_1	m_0	m_5	m_4
m_3	w_3	w_4	w_5	w_0	w_1	w_2	w_3	m_4	m_5	m_0	m_1	m_2	m_3
m_4	w_2	w_5	w_4	w_1	w_0	w_3	w_4	m_1	m_0	m_5	m_4	m_3	m_2
m_5	w_5	w_2	w_1	w_4	w_3	w_0	w_5	m_0	m_1	m_2	m_3	m_4	m_5

(rows list partners from most to least preferred; preferences are fully opposed: woman w ranks man m in position $5 - r$ whenever m ranks w in position r). Its structural profile is extreme in every measured respect: it has *fifteen* rotations, every one a transposition (size 2), for a total move count of 30 — the full budget $n(n - 1)$ of Fact 2 with every man moving the maximal five times, i.e. every man’s trajectory is his entire preference list. The rotation poset on these fifteen transpositions has 24 cover relations and exactly 48 downsets, which our three independent counters confirm as its stable-matching count. In the enumeration, only the shard family containing this schedule reaches 48; regions of the tree using any rotation of size ≥ 3 peak strictly lower.

7 Validation

Ground truth. The same enumerator, parametrized by order, reproduces every known value: order 3 (8 nodes, max 3), order 4 (409 nodes, max 10), order 5 (498,599 nodes, max 16). The last is decisive: $f(5) = 16$ is certified in the companion development [7] by a Lean 4 proof with SAT refutations checked end-to-end by the formally verified checker `cake_lpr` — a machine-checked theorem serving as test oracle for the present method.

Symmetry on/off. Disabling both reductions of Lemma 3 multiplies the tree ($4.5\times$ at order 4; $14\times$ at order 5; $48\times$ in a budget-16 order-6 check of 2,657,773,436 nodes) and never changes a maximum.

Independent implementation. A separately written Python enumerator (different pruning, different code path) agrees on the maxima at the budgets where it is feasible; a third, 40-line reference implementation written directly from Definition 1 and Lemma 3 reproduces the order-3 table of Section 4 node for node. A fourth route bypasses enumeration entirely: a SAT reformulation of the schedule space (bounded frames, first-visit-derived read-off comparisons, selector slots) refutes “some order-5 schedule has $\text{sc}(R(S)) \geq 17$ ” in four minutes, with the refutation checked end-to-end by `drat-trim` and `cake_lpr` — both an independent certificate-backed confirmation at order 5 and the pilot for certifying the order-6 bound itself (see `REPLAY_DESIGN.md` in the repository).

Other search spaces. Before the enumeration, 9.7×10^7 hill-climbing evaluations in raw instance space and 1.2×10^9 evaluations in swap-schedule space (72 million valid schedules) never exceeded 48; an earlier maximal-only pass of the full enumeration (1.40×10^{10} exact counts) also attained 48.

Structural corroboration. All 450 sampled order-5 maximizers share a single rotation-poset skeleton; the dihedral order-6 instance uses the full budget of fifteen size-2 rotations; and across the enumeration only the shard containing the dihedral family reaches 48, with large-rotation shards peaking strictly lower — consistent with extremal instances being full-budget, all-size-2 phenomena.

8 Formalization in Lean 4

Lemmas 1 and 2 are formalized in Lean 4 with Mathlib in the accompanying development ($\sim 4,600$ lines across nine files, from `SixBridge.lean` through `WRelabel6.lean`, atop the order-5 infrastructure of [7]). The end statement is unconditional: for every well-formed order-6 instance I there is a legal schedule S with $\text{sc}(I) \leq \text{sc}(R(S))$ (`validity_unconditional`), obtained by composing the validity chain with a *bottom-agnostic* form of the bridge (`count_le_of_orderPreserving`: J dominates I whenever J ranks each person’s I -stable partners on top in I -order, with no condition on the rest of the lists) and a women-only relabelling that normalizes the man-optimal matching to the identity (`stableCount6_wrelabel6`, `manOpt_wrelabel6`). Lemma 3 is *not* formalized: its relabelling half is proved at the instance level (`Sym6.lean`), its commutation half is validated computationally only, and the certification campaign of Section 9 is designed to need neither (canonical first-appearance cubes are generated without the commutation rule). Every theorem is proved with no `sorry`, and `#print axioms` reports only the standard `propext`, `Classical.choice`, `Quot.sound`; continuous integration rebuilds the development and re-checks these claims on every commit.

Design. Instances are pairs of 6×6 rank tables over `List Nat`; a matching is a list, stability is a decidable predicate, and the set of stable matchings `sms6 I` is computed by filtering the (structurally recursive, hence kernel-computable) permutation enumeration. The formalization uses *no* Gale–Shapley algorithm and *no* rotation machinery: rotations, which carry the classical proofs, never appear. This is a deliberate trade — the classical route (rotations exist, form a poset, covers eliminate single rotations) is textbook material but formalization-heavy, whereas everything the reduction actually needs can be rebuilt from elementary list lemmas.

The bridge lemma. The read-off instance is defined by a promotion operator: `readoff I` moves each person’s stable partners (a decidable predicate) to the top of their list, preserving relative order. The Lean statement of Lemma 2 is a direct case analysis: a would-be blocking pair of a stable matching in the read-off instance either involves a non-trajectory partner (ranked below everything relevant by promotion) or maps back to a blocking pair of the original instance.

The chain. Lemma 1 needs the lattice traversal. Instead of rotation elimination, the development constructs the maximal chain directly:

```
def chainFrom (I : Inst6) : Nat → List Nat → List (List Nat)
| 0, cur => [cur]
| fuel+1, cur => match (sms6 I).filter (strictDomB I cur) with
| [] => [cur]
| c :: cs => cur :: chainFrom I fuel (pickMin I c cs)
```

where `pickMin` selects, among the strict dominators of the current matching, one of *minimum total rank-sum*. This argmin choice does the work of rotation theory for free: any stable matching strictly between the current one and its successor would have a strictly smaller rank-sum than the successor, so every step is automatically a *cover* of the dominance order; the rank-sum strictly increases and is bounded by 30, so fuel 31 provably suffices and the chain ends at the woman-optimal matching. Completeness — every stable pair of I appears on the chain — follows from a *no-skip* lemma (on a cover $\mu \leq \nu$, no man passes over a stable partner) by squeezing each stable pair’s rank between the chain’s endpoints. The endpoint matchings themselves are obtained lattice-style: the meet of two stable matchings is stable (Knuth’s lemma, proved by blocking-pair extraction over lists), a fold of meets gives the man-optimal matching, and the woman-optimal side comes free by a transposition functor that swaps the two sides and inverts matchings.

Trajectories, budgets, cycles. Each clause of Definition 1 and Lemma 1 is an individual theorem about the chain:

clause	Lean theorem(s)
runs from man-optimal to woman-optimal	<code>chainFrom.head</code> , <code>chainFrom.last</code>
all elements stable matchings	<code>chainFrom.mem</code> , <code>chain.mem.perm</code>
consecutive elements are covers	<code>chainFrom.cover</code>
every stable pair realized	<code>chain.complete</code>
men descend their lists, exactly the stable partners	<code>traj.mem.iff</code> , <code>traj.rank.pairwise</code>
from optimal to pessimal partner	<code>traj.head</code> , <code>traj.getLast</code>
no revisits, either side	<code>traj.nodup</code> , <code>wtraj.nodup</code>
≤ 5 moves per man, ≤ 30 total	<code>traj.length.le</code> , <code>total.moves.le.30</code>
steps are disjoint cyclic swaps, size ≥ 2	<code>prevOwner.moved/ne/inj/getD</code>

The last row deserves a remark: the classical statement “every cover eliminates a single rotation” is *not* formalized, because it is not needed. For coverage of the schedule space it suffices that each step’s moved set decomposes into disjoint cycles of size ≥ 2 (the map “previous owner of my new wife” is a fixed-point-free injective self-map of the moved set — four short permutation lemmas); a multi-cycle step splits into consecutive single-cycle steps with identical trajectories.

What remains. The one unformalized link is the exhaustive enumeration itself: a verified replay of the 2.66×10^{10} -node DFS. The per-node evaluation is already expressible — `sms6` of a read-off instance is kernel-computable — so the remaining work is scale engineering (a certified checker consuming an enumeration trace, or a verified reimplement of the search), not new mathematics.

9 Related work and outlook

The computational lineage on $f(n)$ is Eilers’ ($f(4)$, $f(5)$, and the extremal-profile counts, which we replicated independently in [7]), building on Thurber [3] and Knuth [1]; the asymptotic bounds are [4, 5]. Certified computational values of comparable flavour include $R(4, 5) = 25$, the empty hexagon number, Keller’s conjecture and Schur number five; our companion paper [7] follows that tradition for $f(5)$, and Section 8 carries it into the present reduction; a verified replay of the DFS is the remaining step toward end-to-end certification. The next open value, $f(7)$ (best lower bound 71, budget 42), appears to need either a much faster certified evaluator or new structural cuts: the schedule tree grows by roughly three orders of magnitude.

An open problem. The extremal data of Sections 4 and 6 share a pattern. At orders 3 through 6 the maximum is attained by an *all-transposition* schedule (every step of size 2): (01)(02) at order 3; six transpositions (the full budget 12) at order 4; eight transpositions (16 of the budget 20) at order 5; fifteen transpositions (the full budget 30) at order 6. At the *even* orders the extremal schedule exhausts the budget — every man’s trajectory is his entire list — while at the odd orders it provably cannot ($f(5) = 16$ was confirmed here with slack 4). This is consistent with Eilers’ observation that the natural lower-bound constructions look exact only at even orders.

Conjecture 1. *For every $n \geq 3$, $f(n)$ is attained by the read-off of an all-size-2 schedule; for even n , by a full-budget one ($n(n-1)/2$ transposition steps).*

If true, computing $f(n)$ reduces to an optimization over sequences of transpositions — sorting-network-like objects — a space of roughly $(n(n-1)/2)!^{1/2}$ rather than $n!^{2^n}$, and structural insight into it could sharpen the 2.28^n – 3.55^n asymptotic band.

Artifact. All code, logs and verification instructions accompany the repository; the enumeration is reproducible in ~ 10 laptop-hours.

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